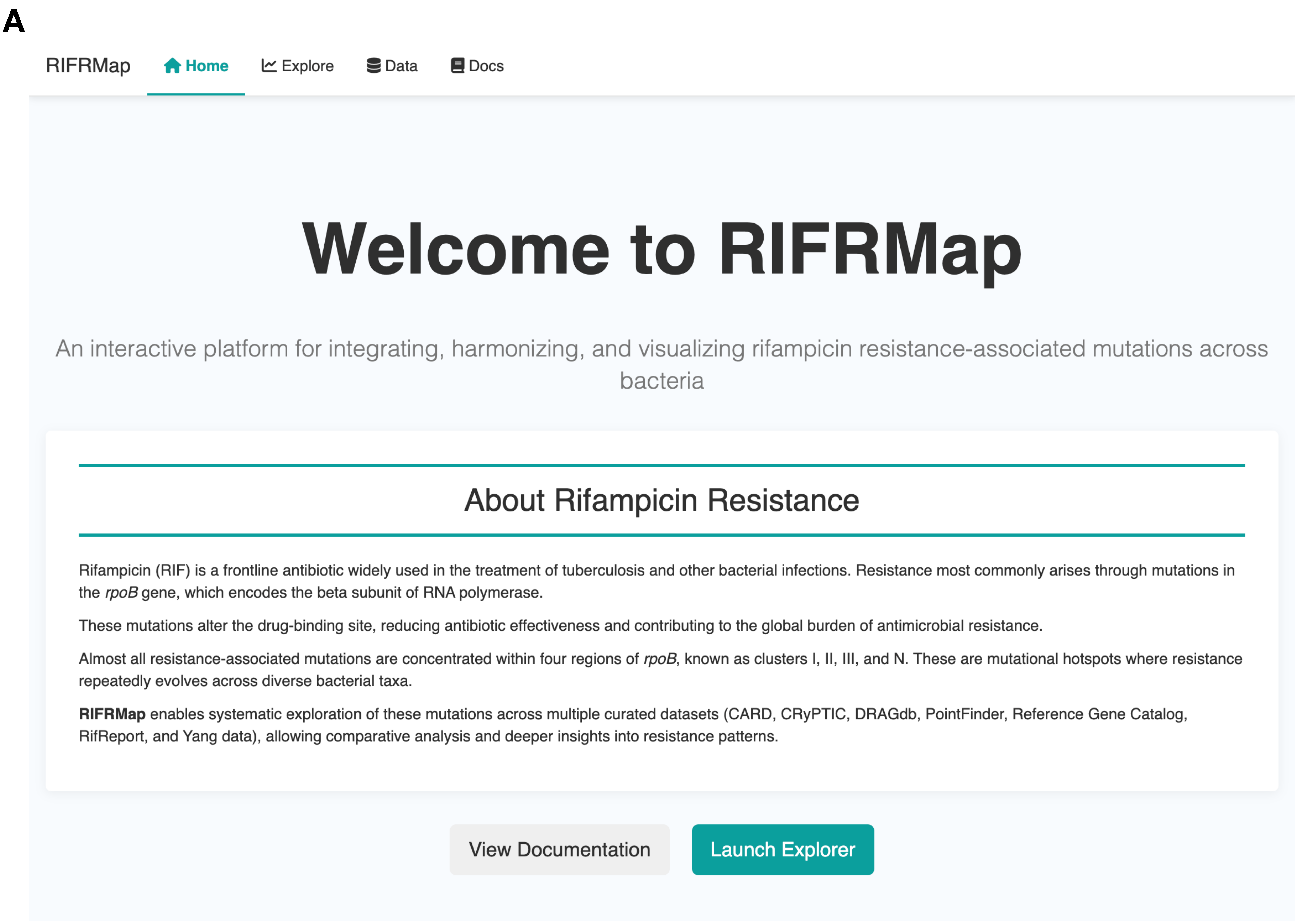


Welcome to RIFRMap

An interactive platform for integrating, harmonizing, and visualizing rifampicin resistance-associated mutations across bacteria

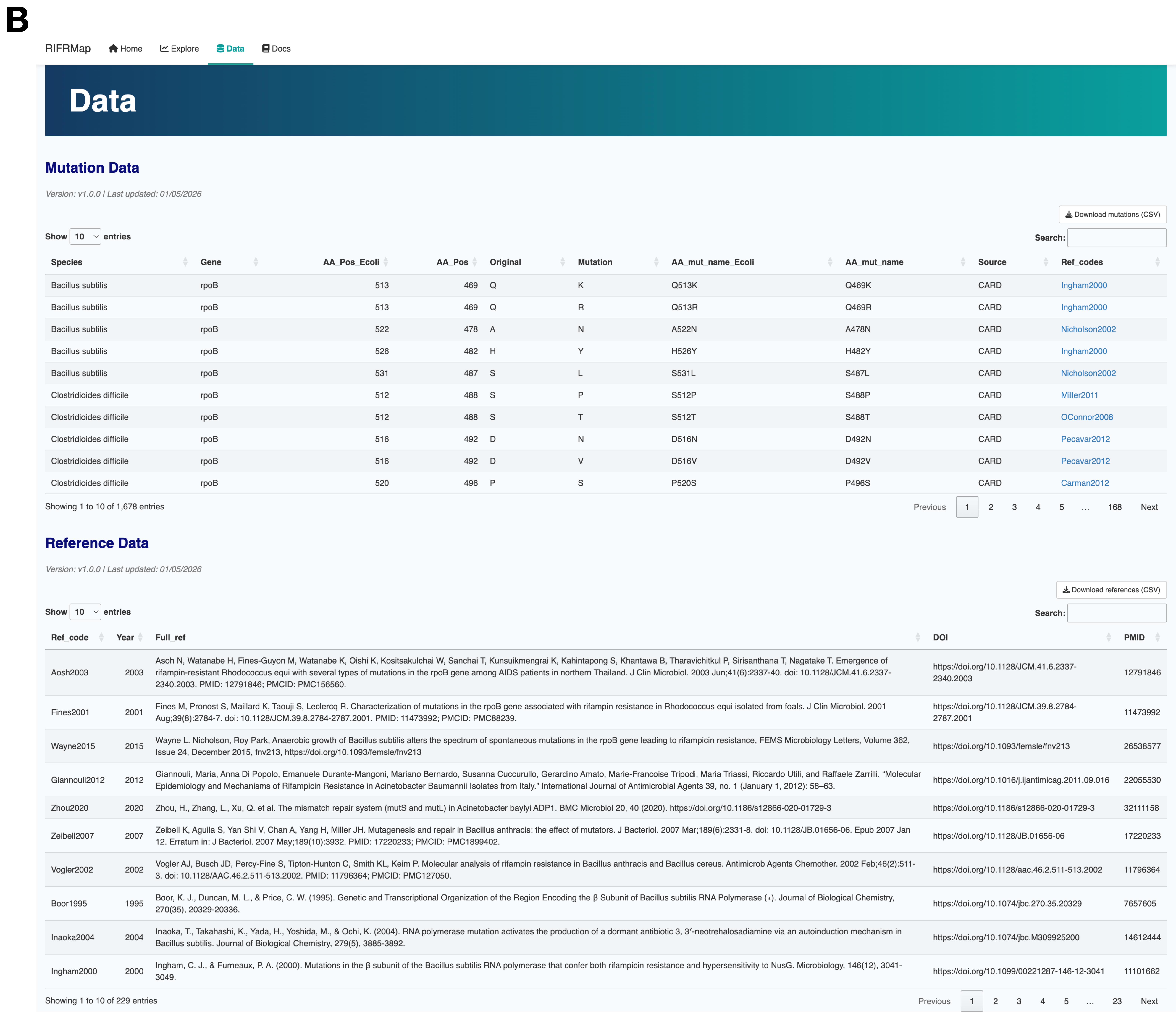
About Rifampicin Resistance

Rifampicin (RIF) is a frontline antibiotic widely used in the treatment of tuberculosis and other bacterial infections. Resistance most commonly arises through mutations in the *rpoB* gene, which encodes the beta subunit of RNA polymerase.

These mutations alter the drug-binding site, reducing antibiotic effectiveness and contributing to the global burden of antimicrobial resistance.

Almost all resistance-associated mutations are concentrated within four regions of *rpoB*, known as clusters I, II, III, and N. These are mutational hotspots where resistance repeatedly evolves across diverse bacterial taxa.

RIFRMap enables systematic exploration of these mutations across multiple curated datasets (CARD, CRyPTIC, DRAGdb, PointFinder, Reference Gene Catalog, RifReport, and Yang data), allowing comparative analysis and deeper insights into resistance patterns.



Data

Mutation Data

Version: v1.0.0 | Last updated: 01/05/2026

Species	Gene	AA_Pos_Ecoli	AA_Pos	Original	Mutation	AA_mut_name_Ecoli	AA_mut_name	Source	Ref_codes
Bacillus subtilis	rpoB	513	469	Q	K	Q513K	Q469K	CARD	Ingham2000
Bacillus subtilis	rpoB	513	469	Q	R	Q513R	Q469R	CARD	Ingham2000
Bacillus subtilis	rpoB	522	478	A	N	A522N	A478N	CARD	Nicholson2002
Bacillus subtilis	rpoB	526	482	H	Y	H526Y	H482Y	CARD	Ingham2000
Bacillus subtilis	rpoB	531	487	S	L	S531L	S487L	CARD	Nicholson2002
Clostridioides difficile	rpoB	512	488	S	P	S512P	S488P	CARD	Miller2011
Clostridioides difficile	rpoB	512	488	S	T	S512T	S488T	CARD	OConnor2008
Clostridioides difficile	rpoB	516	492	D	N	D516N	D492N	CARD	Pecavar2012
Clostridioides difficile	rpoB	516	492	D	V	D516V	D492V	CARD	Pecavar2012
Clostridioides difficile	rpoB	520	496	P	S	P520S	P496S	CARD	Carman2012

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Reference Data

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Ref_code	Year	Full_ref	DOI	PMID
Aosh2003	2003	Asoh N, Watanabe H, Fines-Guyon M, Watanabe K, Oishi K, Kositsakulchai W, Sanchai T, Kunsuikmengrai K, Kahintapong S, Khantawa B, Tharavichitkul P, Sirisanthana T, Nagatake T. Emergence of rifampin-resistant Rhodococcus equi with several types of mutations in the rpoB gene among AIDS patients in northern Thailand. J Clin Microbiol. 2003 Jun;41(6):2337-40. doi: 10.1128/JCM.41.6.2337-2340.2003. PMID: 12791846; PMCID: PMC156560.	https://doi.org/10.1128/JCM.41.6.2337-2340.2003	12791846
Fines2001	2001	Fines M, Pronost S, Maillard K, Taouji S, Leclercq R. Characterization of mutations in the rpoB gene associated with rifampin resistance in Rhodococcus equi isolated from foals. J Clin Microbiol. 2001 Aug;39(8):2784-7. doi: 10.1128/JCM.39.8.2784-2787.2001. PMID: 11473992; PMCID: PMC88239.	https://doi.org/10.1128/JCM.39.8.2784-2787.2001	11473992
Wayne2015	2015	Wayne L. Nicholson, Roy Park, Anaerobic growth of Bacillus subtilis alters the spectrum of spontaneous mutations in the rpoB gene leading to rifampicin resistance, FEMS Microbiology Letters, Volume 362, Issue 24, December 2015, fnv213, https://doi.org/10.1093/femsle/fnv213	https://doi.org/10.1093/femsle/fnv213	26538577
Giannouli2012	2012	Giannouli, Maria, Anna Di Popolo, Emanuele Durante-Mangoni, Mariano Bernardo, Susanna Cuccurullo, Gerardino Amato, Marie-Francoise Tripodi, Maria Triassi, Riccardo Utili, and Raffaele Zarrilli. “Molecular Epidemiology and Mechanisms of Rifampicin Resistance in Acinetobacter Baumannii Isolates from Italy.” International Journal of Antimicrobial Agents 39, no. 1 (January 1, 2012): 58–63.	https://doi.org/10.1016/j.ijantimicag.2011.09.016	22055530
Zhou2020	2020	Zhou, H., Zhang, L., Xu, Q. et al. The mismatch repair system (mutS and mutL) in Acinetobacter baylyi ADP1. BMC Microbiol 20, 40 (2020). https://doi.org/10.1186/s12866-020-01729-3	https://doi.org/10.1186/s12866-020-01729-3	32111158
Zeibell2007	2007	Zeibell K, Aguila S, Yan Shi V, Chan A, Yang H, Miller JH. Mutagenesis and repair in Bacillus anthracis: the effect of mutators. J Bacteriol. 2007 Mar;189(6):2331-8. doi: 10.1128/JB.01656-06. Epub 2007 Jan 12. Erratum in: J Bacteriol. 2007 May;189(10):3932. PMID: 17220233; PMCID: PMC1899402.	https://doi.org/10.1128/JB.01656-06	17220233
Vogler2002	2002	Vogler AJ, Busch JD, Percy-Fine S, Tipton-Hunton C, Smith KL, Keim P. Molecular analysis of rifampin resistance in Bacillus anthracis and Bacillus cereus. Antimicrob Agents Chemother. 2002 Feb;46(2):511-3. doi: 10.1128/AAC.46.2.511-513.2002. PMID: 11796364; PMCID: PMC127050.	https://doi.org/10.1128/aac.46.2.511-513.2002	11796364
Boor1995	1995	Boor, K. J., Duncan, M. L., & Price, C. W. (1995). Genetic and Transcriptional Organization of the Region Encoding the β Subunit of Bacillus subtilis RNA Polymerase (∗). Journal of Biological Chemistry, 270(35), 20329-20336.	https://doi.org/10.1074/jbc.270.35.20329	7657605
Inaoka2004	2004	Inaoka, T., Takahashi, K., Yada, H., Yoshida, M., & Ochi, K. (2004). RNA polymerase mutation activates the production of a dormant antibiotic 3, 3′-neotrehalosadiamine via an autoinduction mechanism in Bacillus subtilis. Journal of Biological Chemistry, 279(5), 3885-3892.	https://doi.org/10.1074/jbc.M309925200	14612444
Ingham2000	2000	Ingham, C. J., & Furneaux, P. A. (2000). Mutations in the β subunit of the Bacillus subtilis RNA polymerase that confer both rifampicin resistance and hypersensitivity to NusG. Microbiology, 146(12), 3041-3049.	https://doi.org/10.1099/00221287-146-12-3041	11101662

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